

Simplex Exponential Periods: Reductions and Open Comparison Problems

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Abstract

This candidate draft separates the unrestricted simplex exponential nonvanishing problem from the controlled higher-dimensional families proved in Paper VI. We record the E-function reduction, a conjugate-real case, fixed-phase effective exceptional sets, and an exact positive-amplitude counterexample to a tempting induction. We formulate the actual-chain visibility and higher-asymptotic-coefficient problems that remain. No unrestricted simplex nonvanishing theorem, general numerical period comparison, or new dimension-wide Factorial Conjecture is proved here.

Candidate research draft; not a numbered paper in the series.

Contents

1	The problem and its normalization	2
2	E-functions and a conjugate-real case	2
3	Fixed-phase exceptional sets are finite, not necessarily empty	3
4	Why positive-amplitude induction fails	3
5	Two distinct higher-dimensional extensions	4
5.1	Ambient-dimension extension	4
5.2	Genuine phase-dimension extension	4
6	A potentially weaker route to factorial nonvanishing	4
7	Audit queue and relation to the series	5

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1 The problem and its normalization

Let

$$\Delta_d = \{t \in \mathbb{R}_{\geq 0}^d : t_1 + \cdots + t_d \leq 1\}, \quad F_Q(z) = \int_{\Delta_d} e^{zQ(t)} dt, \quad Q \in \overline{\mathbb{Q}}[t_1, \dots, t_d].$$

The measure is coordinate Lebesgue measure; $F_Q(0) = 1/d!$. For $d = 0$ the integral is the exponential of the constant phase.

Conjecture 1.1 (Simplex exponential nonvanishing). For every $d \geq 1$ and every polynomial $Q \in \overline{\mathbb{Q}}[t_1, \dots, t_d]$, $F_Q(1) \neq 0$.

Absorbing a nonzero algebraic scale into Q makes this equivalent to $F_Q(\xi) \neq 0$ at every nonzero algebraic ξ . Positivity settles real-valued Q . For complex coefficients positivity alone says nothing about cancellation.

The one-dimensional constant-amplitude result is covered by Paper I [4]. Paper V [5] treats explicit planar comparison systems; Paper VI [6] converts controlled weighted planar periods into simplex integrals in every ambient dimension. Those statements do not imply Conjecture 1.1: the phase obtained by planar slicing still depends on only two active coordinates.

2 E-functions and a conjugate-real case

Proposition 2.1. *For each fixed algebraic polynomial Q , F_Q is an E-function. Its Taylor coefficients and a nonzero linear differential annihilator over $\overline{\mathbb{Q}}(z)$ can be computed from Q .*

Proof. Use the polynomial cube parametrization

$$t_j = u_j \prod_{i < j} (1 - u_i) \quad (1 \leq j \leq d), \quad J(u) = \prod_{i=1}^{d-1} (1 - u_i)^{d-i}.$$

It is a diffeomorphism on the interiors, apart from measure-zero boundary identifications. Thus

$$F_Q(z) = \int_{[0,1]^d} J(u) e^{z\tilde{Q}(u)} du = \sum_{m \geq 0} a_m \frac{z^m}{m!}, \quad a_m = \int_{[0,1]^d} J(u) \tilde{Q}(u)^m du.$$

All a_m lie in one number field. Choose a positive integer A clearing the algebraic coefficient denominators of $\tilde{Q}$. Monomial integration divides by a product of d positive integers bounded by $Cm + C_0$. Therefore

$$A^m \text{lcm}(1, \dots, Cm + C_0)^d a_m$$

is an algebraic integer. These clearing integers, and the common clearing integer for $a_0, \dots, a_m$, grow at most exponentially. Every conjugate of a_m is bounded by an exponential, by integration over the fixed cube. These are the arithmetic E-function bounds.

The integrand is holonomic: $e^{z\tilde{Q}}$ is annihilated by its first-order polynomial connection, and multiplication by J preserves holonomicity. Closure under definite integration, with boundary terms included, gives a univariate holonomic function. Iterated integration over constant endpoints 0, 1 can be carried out by holonomic elimination; see the holonomic integration framework in [3]. Endpoint terms remain holonomic, so an inhomogeneous telescoper can be homogenized. This gives a differential annihilator. Monomial integration computes any required finite list of initial coefficients. No claim that a chosen annihilator has its only finite singularity at zero is made. $\square$

Lemma 2.2 (Algebraic zeros survive coefficient conjugation). *Let F be an E -function with algebraic coefficients. If $F(1) = 0$, then $F^\tau(1) = 0$ for every coefficient embedding τ into $\mathbb{C}$.*

Proof. The proof of Beukers' Proposition 4.1 [1] gives this statement explicitly: every local solution of the minimal differential equation vanishes at the algebraic zero, and the same holds for its coefficient conjugates. The conjugate E -function is one of those solutions. An embedding of its coefficient field extends to $\overline{\mathbb{Q}}$. This is not the unjustified assertion that a field automorphism commutes with an infinite analytic sum. $\square$

Corollary 2.3 (Conjugate-real phases). *Suppose the coefficients of Q lie in a number field K having an embedding $\tau : K \hookrightarrow \mathbb{C}$ for which Q^τ has real coefficients. Then $F_Q(1) \neq 0$.*

Proof. The simplex monomial moments are rational, so coefficient conjugation of the Taylor series gives $(F_Q)^\tau = F_{Q^\tau}$. If $F_Q(1)$ vanished, Lemma 2.2 would contradict $F_{Q^\tau}(1) > 0$. $\square$

This applies in particular when the whole coefficient field has a real embedding, not merely when each coefficient separately has some real conjugate. For a scaled integral, apply it to the coefficients of ξQ together.

3 Fixed-phase exceptional sets are finite, not necessarily empty

Proposition 3.1. *For fixed Q , the set of algebraic zeros of F_Q is finite and can be determined effectively from differential and initial-value data.*

Proof. An algebraic entire E -function is a polynomial. If F_Q is a polynomial, its algebraic zeros are computable and finite since $F_Q(0) = 1/d! \neq 0$. Otherwise apply Adamczewski–Rivoal's algorithm, Theorem 2 of [2], to compute all algebraic arguments where F_Q takes an algebraic value, together with the values; retain those equal to zero. The required annihilator and initial coefficients are supplied by Proposition 2.1. $\square$

The algorithm may be expensive. This proposition neither makes the exceptional set empty nor proves that the specific argument 1 is absent. In particular an ordinary-point lifting statement cannot be applied at an unexamined apparent singularity of an arbitrarily chosen scalar operator.

4 Why positive-amplitude induction fails

Proposition 4.1. *There is a strictly positive rational polynomial p on $[0, 1]$ with $\int_0^1 p(t)e^{5it} dt = 0$.*

Proof. Set $g(t) = t^2(1-t)^2$ and $p = g'' + 25g$. Direct expansion gives

$$p(t) = 25 \left(\left(t - \frac{1}{2} \right)^2 - \frac{1}{100} \right)^2 + \frac{14}{25} > 0.$$

On the other hand,

$$\frac{d}{dt}((g' - 5ig)e^{5it}) = p(t)e^{5it}.$$

Both g and g' vanish at 0 and 1, so the integral is zero. $\square$

This is an exact polynomial twisted-Stokes cancellation. It is not a counterexample to Conjecture 1.1, which has constant amplitude and the standard simplex measure. It does prohibit an induction that replaces the simplex by an arbitrary positive polynomial weight and assumes nonvanishing for that broader class. Integrating out a coordinate often introduces an amplitude; that step must retain its specific algebraic structure.

5 Two distinct higher-dimensional extensions

5.1 Ambient-dimension extension

For a polynomial phase in only $u = t_1, v = t_2$ and $n \geq 3$,

$$\int_{\Delta_{n-1}} e^{Q(t_1, t_2)} dt = \frac{1}{(n-3)!} \int_{\Delta_2} (1-u-v)^{n-3} e^{Q(u, v)} du dv.$$

Here Δ_{n-1} can equivalently be represented by n barycentric coordinates summing to one. Paper VI proves nonvanishing for two explicit families of these weights and phases. One uses standard twisted remainder basis elements and permits every nonzero algebraic scale; the other uses a characteristic-five nonexactness theorem for all exponents.

5.2 Genuine phase-dimension extension

For an arbitrary $Q(t_1, \dots, t_d)$ the interior twisted module can have several constituents. Its relative extension by face systems and the actual simplex class must both be controlled. A larger tensor category or a formal comparison statement by itself does not ensure that this particular scalar period survives. The substantive questions are:

1. Which irreducible interior constituent is detected by the actual simplex integration map modulo the linear face span?
2. Can a scalar relation created at an algebraic parameter be lifted in a closed nonsingular E -function system containing that constituent?
3. When several constituents occur, what prevents cancellation of their contributions to the selected constant-amplitude coordinate?

The elementary irreducible-module criterion in Paper V answers these questions only after irreducibility and nonzero detection have been proved. It is not a replacement for proving them.

6 A potentially weaker route to factorial nonvanishing

For the factorial functional of Paper VI, a flat highest form S^D gives

$$\frac{\mathcal{L}_n(f^m)}{\Gamma(Dm+n)} \longrightarrow F_Q(1), \quad Q = f_{D-1}/D \big|_{S=1}.$$

If this leading coefficient vanishes, no counterexample to the Factorial Conjecture follows. Higher asymptotic coefficients involve polynomial amplitudes multiplying e^Q , with those amplitudes determined by the lower homogeneous layers and the gamma-ratio expansion.

Conjecture 6.1 (Asymptotic visibility target). For every nonzero polynomial with highest homogeneous form S^D , some coefficient in a justified factorial-moment asymptotic expansion is nonzero, or a separately controlled exponentially smaller contribution prevents all positive moments from vanishing.

This is a research target, not a theorem or a claimed formal equivalence. An expansion with all power-series coefficients zero can still have nonzero exponentially small terms. Any proposed proof must control its remainder, not only manipulate a formal expansion.

7 Audit queue and relation to the series

The recovered FC(3) handoffs mention quadratic and cubic triangle nonvanishing drafts. Their complete original proof sources were not part of this integration. In particular, a manuscript claim about arbitrary cubic triangle phases and degenerations should not be promoted on the strength of a context summary. A dedicated audit must inspect the relation-row exclusion, the relevant local lattices, and value specialization. Homogeneous ternary-cubic factorial results are not the same assertion as arbitrary cubic exponential nonvanishing on a triangle.

Papers II–III concern relations among compact affine-line periods, and Paper IV concerns relative functional periods with its stated constant field. They can organize face relations and relative systems, but do not automatically give a numerical theorem for arbitrary simplex chains. The candidate multicolumn workstream addresses another issue, namely actual realization of affine orbit equations, and is likewise separate.

Reproducibility and status. The polynomial identity in Proposition 4.1 and the simplex Jacobians used in Paper VI are checked by `scripts/check_comparison_factorial.py`. Such finite checks do not prove the holonomic or transcendence inputs. This document preserves proved reductions and precise open targets, not a completed proof of Conjecture 1.1.

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